

Bias, ESS, and Debiasing in Massively-Parallel Importance Sampling

An empirical audit of alan's MP-IS, and two early follow-ups

Outline

Why look at this

Does MP-IS actually beat HMC?

A bug hunt

Effective sample size

Jackknife bias correction

Where things stand

Further work

Why look at this

Where this started

A colleague's reaction to alan's massively-parallel importance sampling (MP-IS):
"... probably not going to beat HMC per se, what with the posterior estimates being biased by self-normalization, but mostly seems to do a reasonable job of trying to catch up to it eventually."

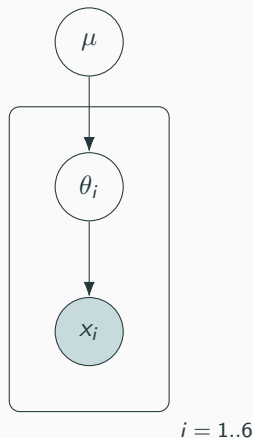
Is that actually true? Let's measure it, on a model small enough that we know the exact answer.

MP-IS in one slide

The idea:

- Draw K candidate values (“particles”) per latent variable, instead of one.
- Reweight them to approximate the true posterior.
- Exploit conditional independence across Plates so this costs $O(K)$ per latent, not $O(K^{\#\text{latents}})$.

No claim of exactness — it’s still importance sampling, just a much more sample-efficient version of it.



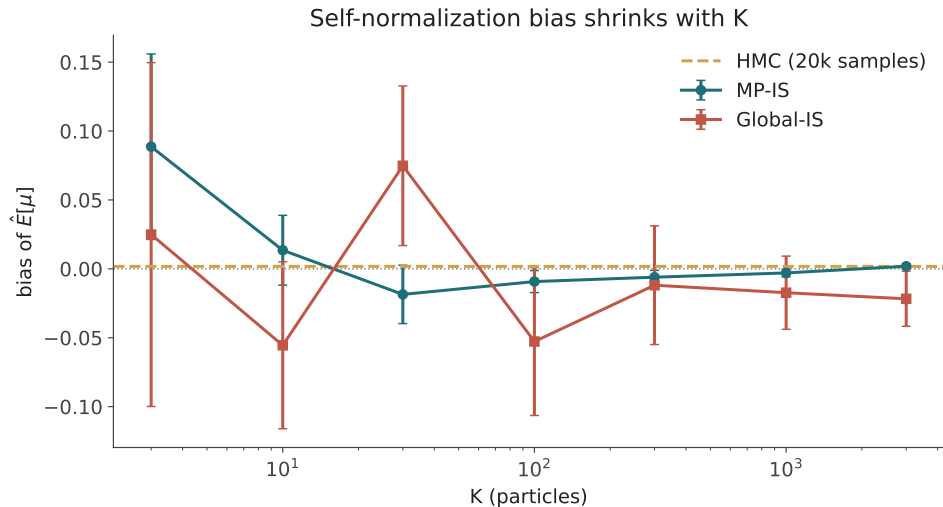
the toy model used throughout this talk

Does MP-IS actually beat HMC?

Setup: a model small enough to know the truth

- $\mu \sim \mathcal{N}(0, 1)$, $\theta_i \mid \mu \sim \mathcal{N}(\mu, 1)$, $x_i \mid \theta_i \sim \mathcal{N}(\theta_i, 0.7)$, $i = 1..6$, x observed.
- Linear-Gaussian $\Rightarrow$ **exact** posterior available (Hessian of the log joint) — no need to trust any single method.
- Compared: alan's MP-IS, plain ("Global") self-normalized IS at the same K , and a hand-rolled HMC sampler validated against the exact posterior.

Headline result



Takeaways

- The colleague was right: MP-IS **is** biased at small K , and the bias **does** shrink and catch up to HMC's own (tiny) bias.
- The sign flip between $K=10$ and $K=30$ is the classic signature of finite-sample self-normalization bias — not just noise.
- Global-IS is consistently 5–10 $\times$ worse than MP-IS at the *same* K across the whole sweep. Plate structure is doing real work.

A bug hunt

A hand-rolled reimplementation didn't match

To dig further into *why* MP-IS is biased, I rebuilt its computation by hand for this toy model, so I could see every intermediate number.

It didn't reproduce alan's own output. Not by a rounding amount — by enough to matter.

Every individual $\log p / \log q$ term matched. The mismatch was in how the pieces get combined.

Two ways to reduce a plate

Order A (what I assumed)

Sum the plate's 6 log-densities together first,
then do one reduction over the shared K particles.

Order B (what alan does)

Reduce over K particles *separately for each of the 6 plate elements*, then sum those 6 already-reduced results.

These are genuinely different computations — the reduction alan uses (log-sum-exp) doesn't distribute over a sum.

Order B is the correct one

- Given μ , the six θ_i 's are **conditionally independent**.
- A sum over a product of independent terms factors exactly into a product of independent sums.
- So each plate element is free to pick whichever of its K particles best explains *its* own observation — a real, quantifiable extra bit of Rao–Blackwellization that plate structure buys you for free.

Rebuilt by hand under Order B: matches alan's ELBO and posterior mean exactly, to float32 precision.

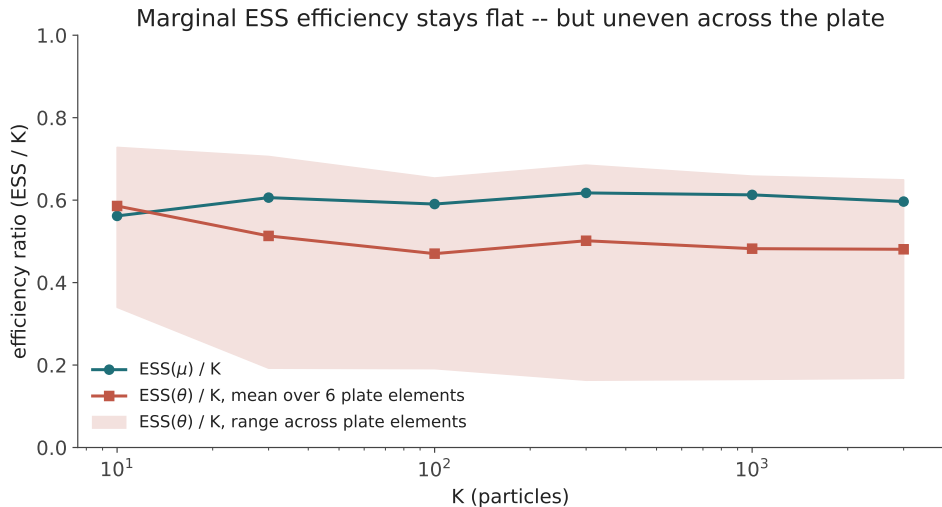
Effective sample size

How many of the K particles are actually doing work?

Kish's effective sample size, for a normalized weight vector:

$$\text{ESS} = \frac{1}{\sum_k w_k^2} \quad (\text{always } \leq K)$$

alan exposes this *per latent variable*, from the marginal posterior it already computes — not something added for this talk.



What the spread means

- Efficiency (ESS/K) for μ stays flat at ≈ 0.6 across the whole range tested — more particles doesn't relatively help or hurt.
- θ 's efficiency is uneven: at $K=3000$, one plate element has $\text{ESS} \approx 504$, another ≈ 1948 — roughly a $4\times$ gap, from the *same* K .
- **Flag for later:** that unevenness turns out to matter a lot for one of the follow-ups at the end of this talk.

Jackknife bias correction

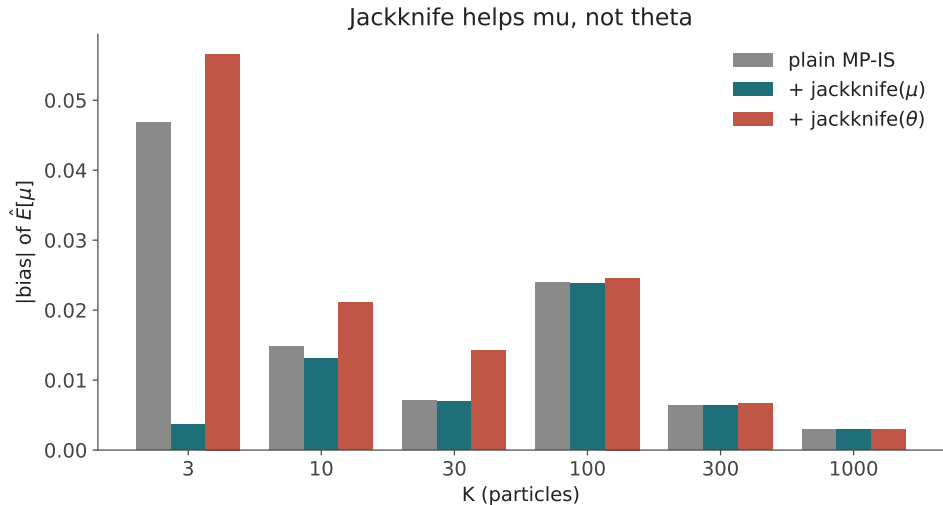
Can we just correct the bias?

Delete-one jackknife, the classical fix for this exact kind of bias:

1. Compute the estimate normally, using all K particles.
2. Recompute it K times, each time leaving one particle out.
3. Extrapolate: $\hat{\theta}_{\text{jack}} = K\hat{\theta} - (K-1)\overline{\hat{\theta}_{-i}}$.

No new machinery — just re-weighted averaging of numbers already computed.

Jackknife results



Why μ but not θ

- μ enters the computation through one flat, linear ratio — exactly the textbook shape jackknife theory is derived for.
- θ enters through a **nested** reduction (Order B, from earlier): a log-sum-exp *inside* another ratio.
- Classical jackknife theory has no mechanism to cancel bias through that extra layer of nonlinearity — and empirically, it doesn't.

A structural rule, not a coincidence: jackknife the *outer*, flat-ratio part of an estimate; don't bother with the nested, per-plate part.

Where things stand

Summary so far

- MP-IS *is* biased at small K , and does converge toward HMC-level accuracy — confirmed, not just assumed.
- The bug hunt uncovered a real, previously undocumented fact about how alan's plate reduction works (Order B).
- ESS is bounded by K and uneven across plate replicates — quantified, not just suspected.
- Jackknife helps, but only for flat/outer reductions — a precise, structural rule for when to bother with it.

Further work

Two follow-up questions

1. Jackknife couldn't fix θ 's bias. Is there a debiasing method that isn't limited to flat ratios?
2. All of the above is still approximate. Can we get an exact posterior out of this same K-particle machinery?

Both are early, toy-model-only prototypes — not yet integrated into `alan` itself. Included here as a look at what's next.

BR-SNIS: debiasing via a Markov chain

Cardoso et al. 2022. Instead of one fixed pool of K samples:

- Keep a *running* candidate: at each step, it joins $K-1$ fresh proposal draws.
- Pick a new running candidate from that pool, weighted by importance weight.
- This defines an exact Markov chain targeting the true posterior for any $K \geq 2$.

Bias is driven down by **chain length**, not by needing $K \rightarrow \infty$ — a fundamentally different lever than jackknife's.

BR-SNIS: what we found

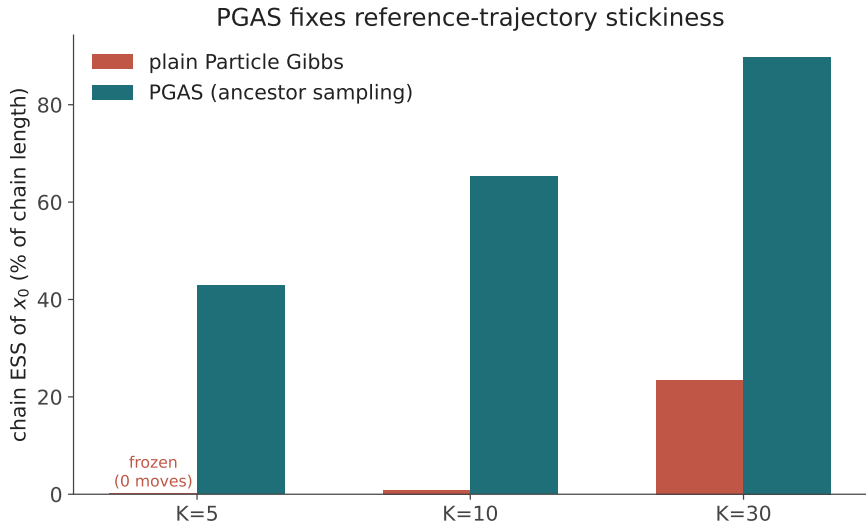
- Run flat, over the whole (μ, θ) joint: **loses to plain MP-IS** — it ignores the plate's conditional independence, so it's $\sim 13\times$ worse on θ at matched compute.
- Run instead on μ 's already plate-reduced ratio (mirroring what jackknife did): competitive with jackknife at matched cost, and **doesn't inherit jackknife's variance blow-up at tiny K** .

Lesson: general-purpose debiasing has to be pointed *at* the structure MP-IS already exploits, not used as a replacement for it.

Particle Gibbs: towards an exact posterior

- MP-IS gives an **unbiased estimator of the marginal likelihood** — exactly the ingredient Particle MCMC needs for exact inference.
- **Particle Gibbs:** pin one full trajectory as a “reference” that survives every resampling step, by construction — the rest of the particles resample normally.
- Needs a genuinely *sequential* model to demonstrate (the toy model above is flat, not sequential) — prototyped instead on a small linear-Gaussian state-space model with a closed-form (Kalman filter) ground truth.

The catch: reference-trajectory stickiness



Why this isn't a surprise

This connects directly back to the ESS finding from earlier: uneven per-replicate weight degeneracy in MP-IS was already a concrete warning sign that plain Particle Gibbs would mix slowly on a model like this.

Ancestor sampling (PGAS) is now treated as a near-term priority, not an optional extra, once real `alan` integration starts.

Where this leaves the design

- Both prototypes are validated on toy models with closed-form ground truth, deliberately *before* touching `alan`'s core sampling code.
- Neither is integrated into `alan` yet — that's the next, more invasive piece of work (`Plate.py` / `Sampler.py`).
- Both findings are written up and cross-linked into the design doc for whoever picks this up next.

Wrap-up

- MP-IS's bias-vs-HMC behavior, measured directly and confirmed.
- A real bug in a hand-rolled comparison led to a genuine finding about how `alan`'s plate reduction works.
- ESS and jackknife give precise, structural (not just empirical) answers about where bias correction does and doesn't help.
- Two early prototypes (BR-SNIS, Particle Gibbs/PGAS) point at what comes next, with concrete numbers already in hand to prioritize the work.

Questions